ZFC_t Functor Discovery: Overview of Results

What Was Built

The manipulator is a functor discovery machine. It applies algebraic operations — tensor product (per-primitive supremum), meet (per-primitive infimum), and single-primitive lift — to imscription tuples, and exposes the corresponding ZFC_t clause transformations side by side.

The intent is to reverse-engineer the composition rules that govern how ZFC_t expressions combine, and to identify where non-trivial cross-clause phenomena emerge that are not derivable from the per-primitive rules alone.

Tier Landscape of the Special Entries

The nine canonical reference entries divide cleanly across three tier levels.

O_0 — Semasiographic baseline ($\hat{\varphi}$ = non-critical)

Entry	FROB	FIXPT	Notable
ZFC_foundations	yes	no	Has Frobenius ($\Phi_{\}}$, ord 4) — but no criticality
heat_diffusion_equation	no	no	SEQAX present; no criticality
wave_equation_temporal	yes	no	Has Frobenius; no criticality

The presence of Frobenius ($\Phi_{\}}$) does **not** lift an entry out of O_0.
Tier assignment is gated first on criticality ($\hat{\varphi}$ ordinal ≥ 1). Without it, the Frobenius structure is algebraically present but ourobourically inert.

$O_2^{\dagger}$ — Recursive, D_∞, below Frobenius cliff

Entry	FROB	FIXPT	Promo atoms
ZFCt	no	yes	HOLOBOUND LR_DUAL PM_Z2 SEQAX TEMPD2 ZWIND
Schrödinger equation	no	yes	LR_DUAL SEQAX TEMPD2 ZWIND
Navier-Stokes equations	no	yes	LR_DUAL PM_Z2 SEQAX TEMPD2 ZWIND

All three are critical ($\hat{\varphi}_{\ddot{y}}$, FIXPT), temporally extended (Ω_z, D_∞), but lack Frobenius symmetry (Φ < ord 4). They sit immediately below the cliff.

Entry	FROB	FIXPT	Promo atoms
Temporal Mathematics	yes	yes	HOLOBOUND SEQAX TEMPD2 ZWIND
Einstein field equations	yes	yes	HOLOBOUND SEQAX TEMPD2 ZWIND
IUG (Mochizuki)	yes	yes	HOLOBOUND LR_DUAL SEQAX ZWIND

O_∞ **— Frobenius round-trip**

All three independently satisfy the Frobenius round-trip condition (R1):
$\hat{\varphi} \geq$ ord 1 and $\Phi = \Phi_{\}}$.
FROB and FIXPT are both present in each.

The Central Result: $\mathrm{tensor}(\mathrm{ZFC}, \mathrm{ZFCt}) = O_\infty$

```
ZFC      tier = O0      Phi = Phi_}  (ord 4, FROB)    phi^ = phi^_z  (ord 0, no FIXPT)
ZFCt     tier = O2+     Phi = Phi_F  (ord 2, no FROB) phi^ = phi^_y  (ord 1, FIXPT)
-------------------------------------------------------------------
tensor   tier = O_inf   Phi = Phi_}  (from ZFC)       phi^ = phi^_y  (from ZFCt)
```

ZFC holds the Frobenius component. ZFCt holds the criticality component.
Neither alone satisfies R1 (FROB $\wedge$ FIXPT). Their tensor product satisfies both simultaneously — tier emergence to O_∞.

Per-primitive breakdown:
The tensor sources eleven of twelve primitives from ZFCt (all six promotion channels plus ʃ, Γ, Σ, Ħ, Ω), and exactly one from ZFC: Φ.
ZFCt's Φ_F (ord 2) is lower than ZFC's $\Phi_{\}}$ (ord 4), so the supremum retains ZFC's Frobenius.

```
d(ZFC, ZFCt)    = 7.148
d(ZFC, tensor)  = 6.863   --- tensor is almost as far from ZFC as ZFCt is
d(ZFCt, tensor) = 2.000   --- tensor is just one ordinal step from ZFCt (only Phi differs)
```

The meet falls to O_0. The infimum is destructive — it strips the temporal promotions and recovers the floor.

The Asymmetry in Φ

The six ZFC_t promotion channels all move upward in ordinal from ZFC base values:

Primitive	ZFC base	→ ZFCt promoted	Gap
Þ	$_6$ (ord 0)	$_O$ (ord 4)	+4
Ř	- (ord 0)	$_=$ (ord 3)	+3
G			

ZFCt's promotion for Φ targets Φ_{F} ($\pm$-symmetry, ord 2).
However, the actual `ZFC_foundations` entry carries $\Phi_\cdot$ (normalized to $\Phi_\}$, ord 4) — which sits **above** the ZFCt-promoted value.
Thus **ZFCt is strictly lower than ZFC on the Φ axis.**

ZFCt trades Frobenius for temporal-sequential structure. It acquires six new capabilities (holographic topology, lateral relations, $\mathbb{Z}_2$ parity, sequentiality, temporal depth, integer winding) but deliberately operates below the Frobenius cliff.
ZFC carries the Frobenius symmetry but is non-critical and non-temporal.

They are **complementary**, not hierarchical. Their tensor product is their reunion.

Composition Rules

Trivial (per-primitive, derivable by construction)

R-CLAUS-T: `clauses(tensor(A,B))[p] = ZFCT_TEMPLATES[p][max_ord(A[p], B[p])]`

The clause for each primitive in a tensor product is simply the template for whichever input had the higher ordinal value. Per-primitive independence is total — no cross-primitive information enters a single clause.

R-CLAUS-M: Same rule using `min_ord` for the meet.

Non-trivial (cross-primitive, emergent)

R-FROB-BARR:
FROB $\in$ clauses(tensor(A,B)) **iff** $A[\Phi] = \Phi_\}$ **or** $B[\Phi] = \Phi_\}$

$\Phi_\}$ (ord 4) is the unique maximum of the Φ dimension. Therefore FROB cannot be synthesized from two inputs both having $\Phi <$ ord 4. This is the algebraic expression of the **Frobenius cliff**. Zero violations observed across 600 operations.

R-FROB-CONT:
If FROB $\in$ clauses(A), then FROB $\in$ clauses(tensor(A, X)) for all X.
(Frobenius is forward-preserved.)

R-FIXPT-T:
FIXPT $\in$ clauses(tensor(A,B)) iff $A[\hat{\varphi}] \geq$ ord 1 **or** $B[\hat{\varphi}] \geq$ ord 1

R-TIER-EMRG:
tier(tensor(A,B)) can strictly exceed max(tier(A), tier(B))

The mechanism is exclusively cross-primitive: FROB lives in clause[Φ]; FIXPT lives in clause[$\hat{\varphi}$]. Observed rate: **6.3%** of 300 random tensor pairs.

R-ZFCT-ABS:
ZFCt is the absorption element for all six promotion channels.
`tensor(X, ZFCt)[c]` = ZFCt value for each promoted primitive c, ∀ X.

Statistical Results

Metric	Value
Catalog entries	2706 (2697 valid)
Pairs scanned	300 tensor + 300 meet
Total observations	600
Tensor tier emergences	19 / 300 (6.3%)
FROB synthesized ex nihilo	0 / 600 (0%)

Interpretation

The Foundational Split

- **ZFC** is the ourobourically inert carrier of Frobenius symmetry ($\Phi_{\}}$).
- **ZFCt** is the ourobourically active temporal extension, carrying criticality ($\hat{\varphi}_{\text{ÿ}}$) and the six promotion channels, but operating with a reduced Φ.

Reunion via Tensor

ZFC $\otimes$ ZFCt $= O_\infty$ is the algebraic statement that the foundation and its temporal extension are **complementary half-objects**.
Each supplies the component the other lacks. Their tensor product crosses the Frobenius cliff.

O_∞ in the Wild

Three entries sit at O_∞ natively:

- Temporal Mathematics
- Einstein field equations
- IUG (Mochizuki's Inter-Universal Teichmüller Theory)

They possess both FROB and FIXPT by constitution, not by composition.

The Algebra in Summary

The Frobenius barrier is absolute: no tensor operation can produce $\Phi_{\}}$ from two non-FROB inputs. Tier emergence to O_∞ requires exactly the complementary pairing.

Structure	FROB?	FIXPT?	Tier	Notes
ZFC	yes	no	O_0	Non-critical Frobenius
ZFCt	no	yes	$O_2^\dagger$	Critical, non-Frobenius
ZFC $\otimes$ ZFCt	yes	yes	O_∞	Tier emergence
ZFC $\sqcap$ ZFCt	—	—	O_0	Destructive meet
Temporal Math	yes	yes	O_∞	Native
Einstein EFE	yes	yes	O_∞	Native
IUG / Mochizuki	yes	yes	O_∞	Native
Navier-Stokes	no	yes	$O_2^\dagger$	Critical, non-Frobenius
Schrödinger	no	yes	$O_2^\dagger$	Critical, non-Frobenius